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EV Charging Stations in Indonesia: Pattern and Opportunity Analysis

An analysis of 3,028 public EV charging stations shows a network that is concentrated in Java and led by PLN, with high-power charging even more concentrated than the stations themselves.

1.1 Context and question

Pasar EV adoption depends on where people can charge. For a business development team, the useful question is not how many chargers exist, but who operates them, where they sit, and how fast they charge.

This analysis asks three things about Indonesia's public charging network:

- Where are the stations, and how concentrated is the network?
- Who runs them, and in what kinds of locations?
- How much of the network offers high-power charging, and where?

1.2 Data and method

The data is the Geocoded Dataset of Public EV Charging Stations in Indonesia (Mendeley Data). It is a single snapshot with no date column, covering 34 provinces. After cleaning, 3,028 stations were usable. Cleaning and preparation, done in Python and Excel, then modeled and visualized in Power BI:

- Removed 1 station record with no province, coordinates or power rating (3,029 became 3,028).
- Read the semicolon-delimited files with the correct encoding and checked column types.
- Grouped provinces into six island groups and power ratings into five tiers, from AC slow (7 kW or less) to DC ultra-rapid (100 kW or more).
- Counted connectors and charger boxes instead of the raw charger count, because the raw field was inflated (one station listed 196 chargers for 16 connectors).
- Built 10 DAX measures, so every percentage in the dashboard is calculated rather than typed in.

1.3 Key findings

1. The network is concentrated in Java. Java holds 2,061 of 3,028 stations (68.1%). The five largest provinces alone hold 2,029 stations (67.0%): DKI Jakarta 731, Jawa Barat 538, Banten 295, Jawa Timur 245 and Jawa Tengah 220.

2. Outside Java the network thins out. Maluku and Papua together have 47 stations (1.6%). The provinces with the fewest stations are Kalimantan Utara (5), Sulawesi Barat, Gorontalo and Papua Barat (8 each), Maluku Utara (9) and Kepulauan Riau (10).

3. PLN leads, but independent operators lead in high-power charging. PLN runs 62.0% of stations. Yet only 9.1% of PLN stations are ultra-rapid (100 kW or more), against 20.7% of independent operator stations.

Provider type	Stations	Share of stations	Ultra-rapid share
PLN / SPKLU network	1,876	62.0%	9.1%
Independent operators	715	23.6%	20.7%
Automaker or dealership	405	13.4%	5.9%

4. High-power charging is more concentrated than the network itself. The country has 348 ultra-rapid stations (11.5% of all). Java holds 281 of them, which is 80.7%, compared with its 68.1% share of all stations. The ultra-rapid share is 13.6% in Java, 9.4% in Sumatera, 7.5% in Bali and Nusa Tenggara, 4.6% in Kalimantan, 3.8% in Sulawesi and 0.0% in Maluku and Papua. The most common tier is AC fast (11 to 22 kW), with 1,670 stations (55.2%).

1.4 Opportunities to investigate

These are hypotheses for a business development team to test, not conclusions. The data shows where stations are, not where drivers need them.

1. High-power charging outside Java. Only 67 of 348 ultra-rapid stations are outside Java. Sumatera, Kalimantan and Sulawesi are the first regions worth checking against EV sales and traffic data, because they already have a network to build on.
2. Partnerships with independent operators. Independent operators build the most high-power stations relative to their size. A company that sells charging equipment or services could study which of their station types and locations perform best.
3. Location types with room to grow. Utility sites (776) and retail and commercial sites (690) host the most stations. Transport corridors have only 169, which may be worth examining for long-distance routes.
4. Provinces with very few stations. Kalimantan Utara, Sulawesi Barat, Gorontalo and Papua Barat have 8 stations or fewer. Whether this reflects a market gap or simply low EV demand is the key question to answer next.

1.5 Limitations and next steps

Limitations

- The dataset is a snapshot with no date column, so it cannot show growth or trends.
- It has no demand or usage data. A province with few stations is not proven to be underserved.

- About 63% of stations have no operator brand recorded, so the operator ranking covers named operators only.
- 21.8% of stations (659) have no location type assigned.
- Power tiers come from rated station power, not from real charging speed.
- The analysis counts stations, not charging capacity, so a station with many connectors counts the same as one with a single connector.

Suggestions steps

1. Add provincial population data from BPS to compare stations per capita.
2. Add EV registration or sales data to compare supply with demand.
3. Collect dated snapshots to measure how fast the network is growing.